

THE OBJECTIVE OF THIS PROJECT IS TO Analyze customer transactions & behavior to identify segments, purchase patterns, and churn risks

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

C:\Users\N yugesh\Downloads\ecommerce_customer_data_custom_ratios.csv

1. LOADING THE DATA

```
df = pd.read_csv("ecommerce_customer_data_custom_ratios.csv")
```

```
df.head()
```

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name
0	46251	2020-09-08 09:38:32	Electronics	12	3	740	Credit Card	37	0.0	Christir Hernandez
1	46251	2022-03-05 12:56:35	Home	468	4	2739	PayPal	37	0.0	Christir Hernandez
2	46251	2022-05-23 18:18:01	Home	288	2	3196	PayPal	37	0.0	Christir Hernandez
3	46251	2020-11-12 13:13:29	Clothing	196	1	3509	PayPal	37	0.0	Christir Hernandez
4	13593	2020-11-27 17:55:11	Home	449	1	3452	Credit Card	49	0.0	Jame Gra

2. Handling Missing and Duplicate values

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          250000 non-null int64
1   Purchase Date                        250000 non-null object
2   Product Category                    250000 non-null object
3   Product Price                       250000 non-null int64
4   Quantity                            250000 non-null int64
5   Total Purchase Amount               250000 non-null int64
6   Payment Method                      250000 non-null object
7   Customer Age                        250000 non-null int64
8   Returns                             202404 non-null float64
9   Customer Name                       250000 non-null object
10  Age                                 250000 non-null int64
11  Gender                             250000 non-null object
12  Churn                              250000 non-null int64
```

```
dtypes: float64(1), int64(7), object(5)
memory usage: 24.8+ MB
```

```
df.isnull().sum()
```

	0
Customer ID	0
Purchase Date	0
Product Category	0
Product Price	0
Quantity	0
Total Purchase Amount	0
Payment Method	0
Customer Age	0
Returns	47596
Customer Name	0
Age	0
Gender	0
Churn	0

```
dtype: int64
```

```
df.dropna(inplace=True)
```

```
df.drop_duplicates(inplace=True)
```

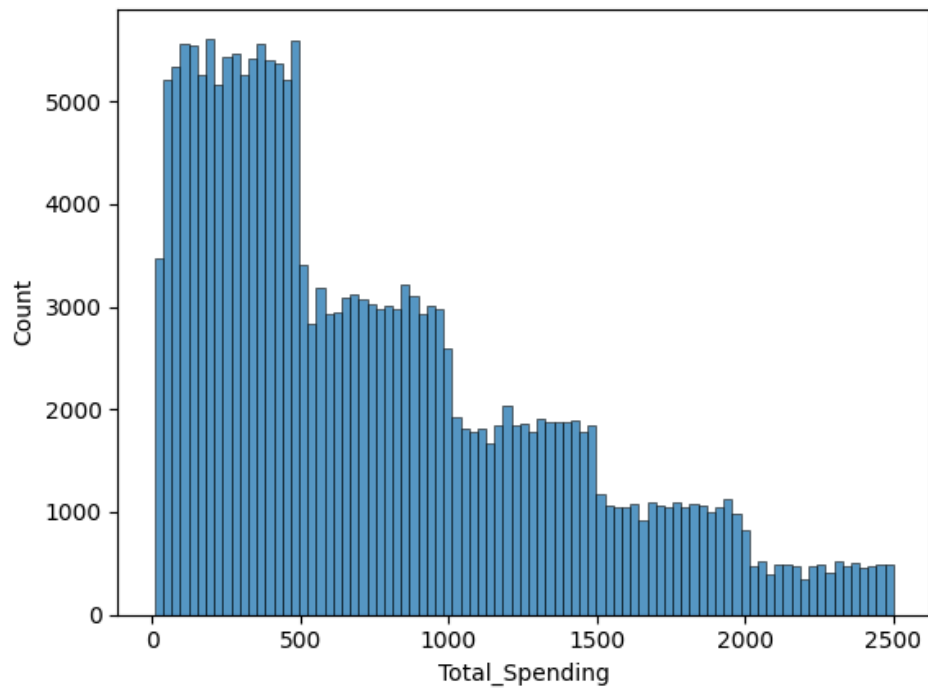
NO DUPLICATES

```
df['Total_Spending'] = df['Quantity'] * df['Product Price']
```

```
df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])
```

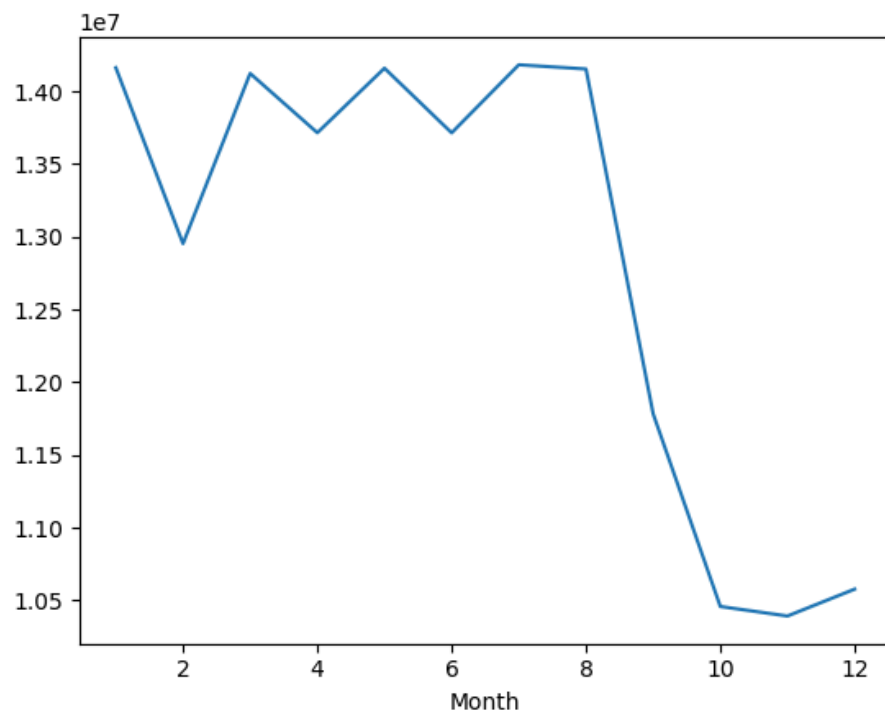
```
df['Month'] = df['Purchase Date'].dt.month
```

```
sns.histplot(df['Total_Spending'])
plt.show()
```



```
monthly_sales = df.groupby('Month')['Total_Spending'].sum()

monthly_sales.plot(kind='line')
plt.show()
```



```
latest_date = df['Purchase Date'].max()

rfm = df.groupby('Customer ID').agg({
    'Purchase Date': lambda x: (latest_date - x.max()).days,
    'Product Category': 'count',
    'Total_Spending': 'sum'
})
```

```
rfm.columns = ['Recency', 'Frequency', 'Monetary']
```

```
rfm.head()
```

	Recency	Frequency	Monetary	
Customer ID				
1	57	1	845	
2	298	2	678	
3	88	7	4591	
4	585	1	276	
5	170	6	3976	

Next steps: [Generate code with rfm](#) [New interactive sheet](#)

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

rfm_scaled = scaler.fit_transform(rfm)
```

```
from sklearn.cluster import KMeans

kmeans = KMeans(n_clusters=4)

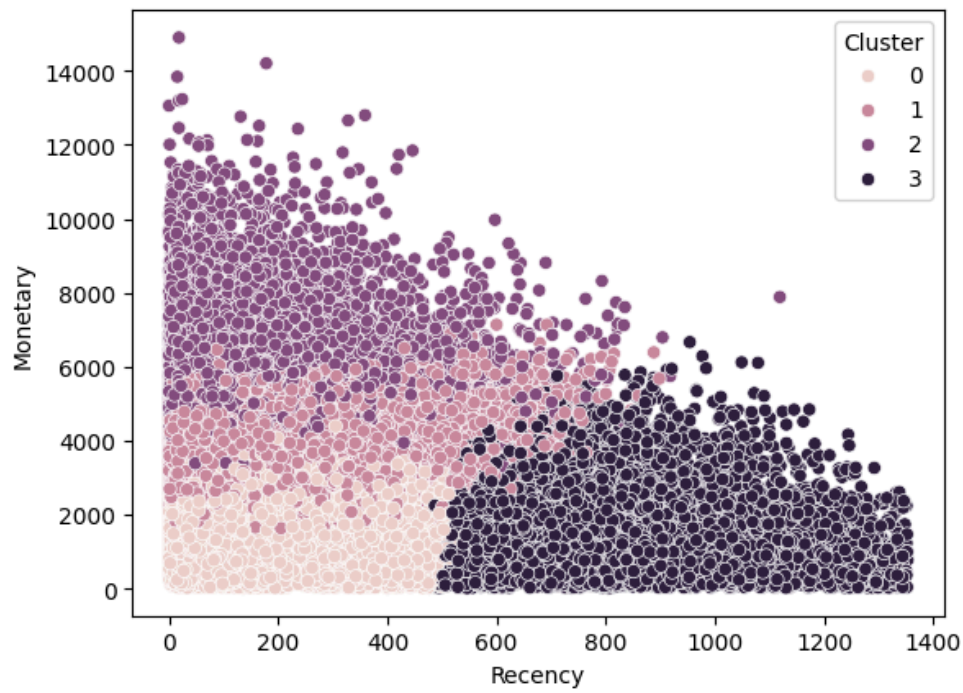
rfm['Cluster'] = kmeans.fit_predict(rfm_scaled)
```

```
rfm.groupby('Cluster').mean()
```

	Recency	Frequency	Monetary	
Cluster				
0	219.903436	2.755764	1725.817447	
1	211.275222	4.842594	3715.216455	
2	158.937111	7.198464	6280.625579	
3	796.420580	2.327468	1702.033679	

```
sns.scatterplot(
    x=rfm['Recency'],
    y=rfm['Monetary'],
    hue=rfm['Cluster']
)

plt.show()
```



```
rfm['Churn_Risk'] = np.where(rfm['Recency'] > 90, "High", "Low")
```

Recomendations

1. Reward Loyal Customers Give discounts and loyalty points.
2. Re-engage Inactive Customers Send email offers.
3. Personalized Recommendations Recommend products based on past purchases